

Integration

Week 6A

Overview

Integration

Key Ideas - *Integration by Simple Substitution Further Functions*

Key Idea - Simple Substitution Further Functions

As we have seen there are occasions when it is possible to perform an apparently difficult integration by first making a substitution to simplify the integrand. This has the effect of changing the variable and the integrand. In this lesson we will meet several other examples of this type, such as

$$\int e^x \sqrt{e^x + 1} \, dx, \quad \int \cos^2 x e^{\sin 2x} \, dx, \quad \int \frac{4x^3 + 2x}{x^4 + x^2} \, dx$$

Be aware that sometimes an apparently sensible substitution does not lead to an integral that you will be able to evaluate. You must then need to try out alternative substitutions.

$$\int f(g(x)) \times g'(x) \, dx$$
$$\int e^x \sqrt{\frac{e^x + 1}{g(x)}} \, dx$$

$\uparrow$ $g'(x)$

Given an integral of the form $\int f(g(x))g'(x) dx$

Step 1: Let $u = g(x)$

Step 2: Find $\frac{du}{dx} = g'(x)$, so $du = g'(x) dx$.

[You may have to adjust constant factors]

Step 3: Make the substitutions $u = g(x)$ and $du = g'(x) dx$ in the original integral, to get
$$\int f(u) du.$$

The integral must now be entirely in terms of u , including the ' dx '.

Step 4: Evaluate the new integral (with respect to u) $\int f(u) du$.

Step 5: In the answer back substitute each occurrence of variable u by $g(x)$.
The answer must now be entirely in terms of x , or the original variable.

Example: Integrate $\int e^x \sqrt{e^x + 1} \, dx$.

In this example we make the substitution $u = e^x + 1$, in order to simplify the square root term.

Step 1: Let $u = e^x + 1$

We shall see that the rest of the integrand, $e^x \, dx$, will be taken care of automatically in the substitution process, and that this is because e^x is the derivative of that part of the integrand used in the substitution, i.e. $u = e^x + 1$

Step 2: $\frac{du}{dx} = e^x$ and $du = e^x \, dx$

So, substitute u for $e^x + 1$, and with $e^x \, dx = du$ in $\int e^x \sqrt{e^x + 1} \, dx$

$$\begin{aligned} \text{Step 3: } \int e^x \sqrt{e^x + 1} \, dx &= \int \sqrt{e^x + 1} e^x \, dx \\ &= \int \sqrt{u} \, du \end{aligned}$$

Now integrate the new function with respect to u .

$$\begin{aligned}\text{Step 4: } \int \sqrt{u} \, du &= \int u^{1/2} \, du \\ &= \frac{u^{3/2}}{3/2} + c \\ &= \frac{2u^{3/2}}{3} + c\end{aligned}$$

We now back substitute each occurrence of variable u by $e^x + 1$, giving

$$\text{Step 5: } \frac{2(e^x + 1)^{3/2}}{3} + c \text{ or } \frac{2}{3} \sqrt{(e^x + 1)^3} + c$$

Note: Check this solution by differentiation $\frac{d}{dx} \left(\frac{2}{3} (e^x + 1)^{3/2} \right)$

Example: Integrate $\int 3 \cos 2x e^{\sin 2x} dx$

In this example we make the substitution $u = \sin 2x$, in order to simplify the square root term.

Step 1: Let $u = \sin 2x$

We shall see that the rest of the integrand, $3 \cos 2x dx$, will be taken care of automatically in the substitution process, and that this is because $\sin 2x$ is the derivative of that part of the integrand used in the substitution, i.e. $u = \sin 2x$

Step 2: $\frac{du}{dx} = 2 \cos 2x$ and $\frac{du}{2} = \frac{2 \cos 2x dx}{2} \Rightarrow \frac{du}{2} = \cos 2x dx$

need (2)

So, substitute u for $\sin 2x$, and with $\cos 2x dx = \frac{du}{2}$ in $\int 3 \cos 2x e^{\sin 2x} dx$

have (3)

Step 3: $\int 3 \cos 2x e^{\sin 2x} dx = 3 \int \cos 2x e^{\sin 2x} dx$

$$= 3 \int e^u \frac{du}{2}$$

$$= \frac{3}{2} \int e^u du$$

Now integrate the new function with respect to u .

$$\begin{aligned}\text{Step 4: } \frac{3}{2} \int e^u du &= \frac{3}{2} e^u + c \\ &= \frac{3}{2} e^{\sin 2x} + c\end{aligned}$$

We now back substitute each occurrence of variable u by $\sin 2x$, giving

$$\text{Step 5: } \frac{3}{2} e^{\sin 2x} + c$$

Note: Check this solution by differentiation $\frac{d}{dx} \left(\frac{3}{2} e^{\sin 2x} + c \right)$

Example: Integrate $\int \frac{-2 \sin 2x}{\cos 2x + 3} dx$

This is a special case of a commonly occurring form,

$\frac{\text{derivative of denominator}}{\text{denominator}}$

where the integral is the **logarithm** of the denominator.

In this example we make the substitution $u = \cos 2x + 3$, in order to simplify.

Step 1: Let $u = \cos 2x + 3$

We shall see that the rest of the integrand, $-2 \sin 2x \, dx$, will be taken care of automatically in the substitution process, and that this is because $-2 \sin 2x$ is the derivative of that part of the integrand used in the substitution, i.e. $u = \cos 2x + 3$.

Step 2: $\frac{du}{dx} = -2 \sin 2x$ and $du = -2 \sin 2x \, dx$

So, substitute u for $\cos 2x + 3$, and with $du = -2 \sin 2x \, dx$ in $\int \frac{-2 \sin 2x}{\cos 2x + 3} dx$

$$\text{Step 3: } \int \frac{-2 \sin 2x}{\cos 2x + 3} dx = \int \frac{1}{u} du$$

Now integrate the new function with respect to u .

$$\text{Step 4: } \int \frac{1}{u} du = \ln|u| + c$$

We now back substitute each occurrence of variable u by $\cos 2x + 3$, giving

$$\text{Step 5: } = \ln|\cos 2x + 3| + c$$

Note: Check this solution by differentiation $\frac{d}{dx} (\ln|\cos 2x + 3|)$

Example: Integrate $\int \cos x \sin^6 x \, dx$.

In this example we first rewrite $\int \cos x \sin^6 x \, dx = \int \cos x (\sin x)^6 \, dx$.

Then we make the substitution $u = \sin x$, in order to simplify the square root term.

Step 1: Let $u = \sin x$

We shall see that the rest of the integrand, $\cos x \, dx$, will be taken care of automatically in the substitution process, and that this is because $\cos x$ is the derivative of that part of the integrand used in the substitution, i.e. $u = \sin x$

Step 2: $\frac{du}{dx} = \cos x$ and $du = \cos x \, dx$

So, substitute u for $\sin x$, and with $\cos x \, dx = du$ in $\int \cos x \sin^6 x \, dx$

Step 3: $\int \cos x \sin^6 x \, dx = \int \cos x (\sin x)^6 \, dx$
 $= \int u^6 \, du$

Now integrate the new function with respect to u .

$$\text{Step 4: } \int u^6 du = \frac{u^7}{7} + c$$

We now back substitute each occurrence of variable u by $\sin x$, giving

$$\text{Step 5: } \frac{(\sin x)^7}{7} + c$$

Note: Check this solution by differentiation $\frac{d}{dx} \left(\frac{(\sin x)^7}{7} + c \right)$

Exercise: Find the following:

(a) $\int \sin x \sqrt{\cos x} \, dx = \int \sin x (\cos x)^{\frac{1}{2}} \, dx$

↓ wrong sub

1. let $u = \cos x$ or $u = \sin x$ $\frac{du}{dx} = \cos x$
 $du = \cos x \, dx$

2. $\frac{du}{dx} = -\sin x$

$du = -\sin x \, dx \Rightarrow \frac{du}{-1} = \sin x \, dx$

3. $\int \sin x (\cos x)^{\frac{1}{2}} \, dx$

$= \int u^{\frac{1}{2}} \frac{du}{-1}$

[integrate]

A. $-1 \int u^{\frac{1}{2}} \, du = -\frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$
 $= -\frac{2u^{3/2}}{3} + C$

S. $-\frac{2(\cos x)^{3/2}}{3} + C = -\frac{2\sqrt{\cos^3 x}}{3} + C$ [back sub]

$$(b) \int x \cos x^2 e^{\sin x^2} dx$$

$$1. \text{ let } u = \sin x^2$$

$$2. \frac{du}{dx} = 2x \cos x^2 \Rightarrow$$

$$3. \int x \cos x^2 e^{\sin x^2} dx$$

$$\frac{du}{2} = \frac{2x \cos x^2 dx}{2}$$

$$= \int e^u \frac{du}{2} = \frac{1}{2} \int e^u du$$

$$4. \frac{1}{2} \int e^u du = \frac{1}{2} e^u + C$$

$$5. \frac{1}{2} e^{\sin x^2} + C$$

$$\therefore \frac{du}{2} = x \cos x^2 dx$$

Slightly harder b!

$$(c) \int \frac{4e^{2x} + e^x}{2e^{2x} + e^x + 1} dx$$

$$\int \frac{2e^{2x} + e^x}{2e^{2x} + e^x + 1} dx$$

1. let $u = 2e^{2x} + e^x + 1$

2. $\frac{du}{dx} = 4e^{2x} + e^x \Rightarrow du = (4e^{2x} + e^x) dx$

3. $\int \frac{1}{u} du = \int \frac{du}{u}$

4. $\ln|u| + c$

5. $\ln|2e^{2x} + e^x + 1| + c$

$$u = 2e^{2x} + e^x + 1$$

$$\frac{du}{dx} = 4e^{2x} + e^x$$

$$\frac{du}{2} = \frac{4e^{2x} + e^x dx}{2}$$

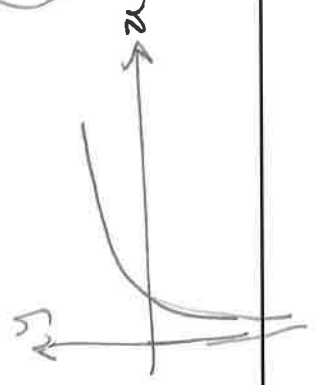
$$= (2e^{2x} + e^x) dx$$

$$\int \frac{1}{u} \frac{du}{2}$$

$$\frac{1}{2} \int \frac{1}{u} du$$

$$\frac{1}{2} \ln|u| + c$$

$$\frac{1}{2} \ln|2e^{2x} + e^x + 1| + c$$



$$(d) \int \tan 2x \sec^2 2x \, dx = \int \tan 2x (\sec^2 2x)^2 \, dx$$

$$\text{let } u = \tan 2x \quad \text{or}$$

$$\frac{du}{dx} = 2 \sec^2 2x$$

$$\frac{du}{2} = \frac{2 \sec^2 2x \, dx}{2}$$

$$\frac{du}{2} = \sec^2 2x \, dx$$

$$\int \tan 2x \sec^2 2x \, dx$$

$$\begin{aligned} & \int u \frac{du}{2} \\ &= \frac{1}{2} \int u \, du \\ &= \frac{1}{2} \frac{u^2}{2} + C \\ &= \frac{u^2}{4} + C \end{aligned}$$

$$= \frac{\tan^2 2x}{4} + C$$

~~$$u = \sec 2x$$~~

~~$$\frac{du}{dx} = \sec 2x \csc 2x$$~~